Submission Summary

Conference Name	International Conference on “Quantum Innovations for Computing and Knowledge Systems (QUICK’26)”
Track Name	QUICK 2026 - Review Papers of All Tracks
Paper ID	110
Paper Title	AUTO-IMMUNE AI–POWERED SECURE DATA MANAGEMENT FOR EMRs WITH FINE-GRAINED ACCESS CONTROL AND RE-ENCRYPTION
Abstract	Abstract—Electronic Medical Records (EMRs) and wearable sensor health data are essential for patient care but highly sensitive. They are vulnerable to unauthorized access, data breaches, and cyberattacks, especially when stored or shared through cloud environments. Existing security mechanisms often rely on weak encryption or lack robust access control, leading to privacy violations and data integrity issues. To address these challenges, an AI DataLock framework was proposed. This system integrates Hierarchical Key–Attribute-Based Encryption (HE-ABE) for fine-grained access control and blockchain technology to provide an immutable, decentralized ledger for tamperproof auditing and transparent data transaction management. HE-ABE ensures that only users with the correct attributes can access the patient data. A key innovation is the real-time attack detection mechanism powered by the Isolation Forest algorithm, which identifies anomalous access patterns and potential intrusions with high accuracy. Upon detecting a threat, the system activates a dynamic re-encryption process using Proxy Re-Encryption (PRE). This approach securely re-encrypts the affected health records with fresh cryptographic keys without exposing the underlying plaintext, rendering previously captured ciphertext useless to attackers. The proposed system ensures the confidentiality, integrity, and accountability of healthcare data stored in the cloud. Index Terms—Electronic Medical Record (EMR), Hierarchical Key-Attribute-Based Encryption (HE-ABE), Blockchain, Isolation

	Forest, Proxy Re-Encryption (PRE), Auto-Immune Security, Fine-Grained Access Control.
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Primary Subject Area	Multidisciplinary Engineering and Technology
Secondary Subject Areas	Embedded System and Soft Computing
Submission Files	Auto_immune_AI.pdf (634.4 Kb, 12/16/2025, 9:45:23 AM)
Supplementary Files	Supplementary Material.pdf (430.9 Kb, 12/16/2025, 9:58:12 AM)